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Surrogate Selection for Reinforcement-Learning HVAC Control: Step Size, Not Predictive Accuracy, Predicts Control Utility

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ABSTRACT Reinforcement-learning (RL) controllers for building HVAC systems are trained against fast learned surrogates of the building, and surrogates are selected by predictive error. This paper shows that predictive error is the wrong selection criterion and identifies one that works. On the BOPTEST benchmark a calibrated grey-box twin predicts far better (24 h rollout error $0.644\,^{\circ}\mathrm{C}$ against $1.557\,^{\circ}\mathrm{C}$ for a compact black-box surrogate) yet trains a controller that fails on every seed, with comfort violation above 77%; retraining the same architecture at finer resolution makes it more accurate ($0.876\,^{\circ}\mathrm{C}$) and equally unusable. That retraining also inverts the model's response to both control inputs, reproducibly and invisibly to the predictive metric. A rescaling control that holds scale-free response roughness fixed while varying the integration step isolates the operative variable: the magnitude of the surrogate's per-step increment, measurable on the surrogate alone before any policy is trained. We then introduce role-separated surrogate training, which stops trading smoothness against fidelity by assigning the two requirements to different models: a black-box surrogate supplies the rollout dynamics while a frozen physics-informed twin acts only as a per-step model-disagreement reward censor. It gives the best worst-window maintenance score of the study (0.041 typical) at 85 times live-simulator throughput, against 0.547/0.869 for a receding-horizon MPC baseline and 0.910 for the emulator's own controller. The planner reproduces the same inverted ordering, so the effect belongs to optimizing through the surrogate, not to reinforcement learning. Of five pre-specified hypotheses, three are falsified, including two of our own.

INDEX TERMS Air conditioning, buildings, computational modeling, digital twins, predictive models, reinforcement learning.

I. INTRODUCTION

TRAINING a reinforcement-learning (RL) controller requires millions of environment interactions, which no real building and no high-fidelity Modelica simulator can supply. The standard response, across essentially the whole HVAC RL literature [1], [2], is to fit a fast learned surrogate of the building and train the policy against it [3], [4]. That makes a design decision unavoidable: among candidate surrogates, which one should be used as the training environment?

In practice the decision is made on predictive error. Surrogate papers report multi-step rollout error, error CDFs and calibration statistics, and the surrogate with the lowest error is the one carried forward [5]–[9]. This is a reasonable-looking heuristic and it is, to our knowledge, universal in the building-energy literature. It is also untested: predictive error is measured on held-out trajectories, whereas a training environment is used as a closed-loop transition function on *policy-induced* states, and nothing guarantees that the two orderings agree.

They do not. On the BOPTEST benchmark we find the ordering is not merely uninformative but inverted over part of the range: the *least* accurate surrogate in the study is the only one that trains a usable controller, and the most accurate one trains a controller that fails outright. A calibrated grey-box twin reaches a 24 h rollout error of $0.644\,^{\circ}\mathrm{C}$ against $1.557\,^{\circ}\mathrm{C}$ for a compact black-box model, yet a policy trained on it leaves the comfort band for more than 77% of the horizon on every random seed, while the coarse black-box model holds it for all but a few per cent of the time. Retraining the *same* black-box architecture on a finer corpus makes it strictly more accurate ($0.876\,^{\circ}\mathrm{C}$) and equally unusable, which rules out model class as the explanation.

That retraining turns out to break something the predictive metric does not measure. The finer-corpus surrogate inverts its response to the supply-temperature command in about nine sampled states out of ten, so that a hotter command predicts a colder zone, reproducibly across four independent training draws, while its 24 h rollout error improves in every one of them. The fan command is inverted too. Constraining training to preserve the supply-response sign restores that channel to 100% at a *lower* rollout error still ($0.745\,^{\circ}\mathrm{C}$) but leaves the fan channel inverted, and the controller trained on the repaired surrogate exploits the remaining defect instead. Predictive error therefore hides defects that a controller will find, on every actuated input, and repairing one input is not enough. Because of this the matched-resolution backend is reported as a failure of the predictive criterion rather than as evidence for the mechanism, which rests instead on the calibrated twin and on a rescaling control that preserves response signs by construction.

The question this paper answers is therefore not whether such a failure can occur, but *what property of a surrogate does predict training utility*, given that predictive error does not. We identify it as the magnitude of the surrogate's per-step state increment, that is its effective step size, and we isolate it causally rather than by correlation: a rescaling control that leaves the scale-free roughness of the action-response surface unchanged while changing only the per-step increment reproduces the collapse. The property is measurable on the surrogate alone, before any controller is trained, which turns it into an actionable selection criterion rather than a post-hoc diagnosis.

Identifying the criterion also shows why the usual remedy fails. Smoothness and physical fidelity are in tension in a single model: coarsening the time step buys the optimization-friendly response surface that policy-gradient search needs, and pays for it in predictive accuracy. We resolve the tension by refusing the trade-off, assigning the two requirements to two different models rather than compromising within one. A compact black-box surrogate supplies the rollout dynamics, and an inverse-calibrated physics-informed twin is restricted to a *frozen, forward-only reward censor* that penalizes per-step disagreement between the two. The twin never enters the rollout and never enters the policy loss, so policy search still runs over the smooth dynamics that make it tractable while state–action regions the physics disputes are discouraged.

Contributions. This paper makes three methodological contributions.

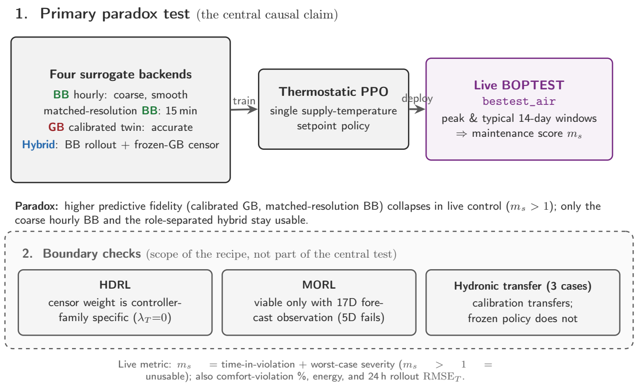

FIGURE 1. Study design. *Top, the selection-criterion test:* four surrogate backends spanning a range of predictive accuracy and per-step increment (coarse hourly BB, matched-resolution BB, calibrated GB, and the role-separated hybrid) each train the same thermostatic PPO controller, which is then scored zero-shot on the live BOPTEST runtime. Predictive accuracy and control utility order the backends differently: the two most accurate single-model backends collapse ($m_s > 1$). *Bottom, boundary tests (dashed box):* hierarchical DRL, multi-objective RL and a three-case hydronic transfer study map where the recipe holds and are not part of the central test.

1) **A surrogate-selection criterion for RL training environments.** We show that predictive error fails as a predictor of downstream control utility, and that the per-step increment magnitude succeeds. The criterion is controller-independent, measurable on the surrogate in isolation, and established with a rescaling control that separates it from response-surface smoothness, with which it is otherwise confounded. We further show that predictive error misses a second and simpler defect, the loss of the control-response sign on each actuated input independently, and that this too is detectable before a controller is trained.
2) **Role-separated surrogate training.** We formulate a training architecture that satisfies the smoothness and fidelity requirements simultaneously instead of trading them off, by assigning them to a rollout model and a frozen disagreement censor respectively. On the benchmark it recovers the best worst-window robustness of the study at 85 times live throughput, and two negative controls establish that the gain comes from the role assignment and not from model averaging or from better initialization.
3) **A falsification protocol for surrogate-based RL claims.** Four hypotheses are pre-specified before the corresponding runs and reported whether supported or not, with seed replication on the central test and on the censor-weight sweep. Two of the four are falsified, including one of our own, which is what allows the negative outcomes to be read as findings rather than as tuning artifacts.

The pre-specified hypotheses are:

H1. *Fidelity–utility:* a surrogate with higher predictive fidelity is a better RL training environment.

H2. *Role separation:* using the physical twin as a frozen reward-shaping censor over a smooth black-box rollout recovers the control utility that direct use of the twin destroys.

- **H3.** *Censor-weight universality:* one censor weight for model disagreement transfers across controller families.
- **H4.** *Transferability:* the inverse-calibration pipeline and the frozen controller transfer to related hydronic test cases under documented recalibration.
- **H5.** *Controller-class specificity:* the fidelity–utility inversion is a property of policy-gradient learning, so a receding-horizon planner using the same surrogates escapes it. Registered after the first four, once the inversion itself had been established, and reported here whether favorable or not.

Evidence is kept deliberately narrow (Fig. 1). The primary causal test is a four-backend comparison under one thermostatic PPO controller; hierarchical DRL, multi-objective RL and a three-case hydronic transfer study are boundary tests that map where the recipe holds, not additional support for the central claim. Section II positions the work, Section III defines the control problem and the architecture, Section IV reports and discusses the results, and Section V concludes. Reproduction detail is in the supplementary material.

II. RELATED WORK

A. BENCHMARKS AND HVAC DEEP REINFORCEMENT LEARNING

Reproducible HVAC-control research depends on a shared benchmark, because controller claims otherwise vary with the simulator, weather year, disturbance schedule and comfort definition. BOPTEST serves Modelica building models through a uniform API [10], [11] with a Gym interface for RL workflows [12], complementing demand-response benchmarks such as CityLearn [13], [14]; every controller here is finally re-scored in BOPTEST, not only in its learned environment. Against this benchmark, HVAC DRL studies compare learned policies with rule-based, Guideline 36 or MPC references [2], [15]–[17] and extend them to multi-agent and context-aware settings [18], [19]; hierarchical policies decompose comfort and energy objectives across timescales [20]–[22], and multi-objective RL represents the energy–comfort trade-off explicitly [23]–[26]. These works fix the training environment, which is precisely what this study varies.

B. LEARNED BUILDING SURROGATES

Learned building surrogates make DRL-scale training feasible where direct simulation is too slow, spanning black-box networks, grey-box RC models, Neural ODEs and physics-informed learning [5]–[9], [27]–[30], and connecting to the broader scientific-machine-learning literature on operator learning and projection-based reduced-order models [31]–[33]. Surrogate development has itself been framed as a staged validation protocol [3], and fast-surrogate work emphasizes runtime feasibility [4]. Most of this literature validates prediction metrics, yet a controller, like a surrogate-based optimizer that can drive an accurate surrogate into infeasible optima by exploiting its residual error [34], uses the surrogate as a closed-loop transition function under policy-induced states. We therefore report predictive fidelity and runtime first, then test the stronger claim that fidelity implies training utility.

C. OBJECTIVE MISMATCH AND DISTRIBUTION SHIFT

That stronger claim is a distribution-shift question. On-policy data collection and observation design shape the states an RL controller learns on [35]–[37], and safe-RL work shows that small model errors become operationally important once actions are optimized against them [16], [38]. This links HVAC control to model exploitation and dataset shift in model-based RL [39], [40], and specifically to the *objective mismatch* between a surrogate trained for predictive likelihood and one that yields a good downstream policy [41], [42]; decision- and value-aware model learning answer this mismatch by retraining the model toward the control objective [43], and the same low-error-yet-poor-transfer effect is central to sim-to-real deployment [44]. Training a policy inside a learned model is the model-based, or Dyna, tradition [45], now scaled to world-model agents [46]. Unlike value-aware retraining, we hold both surrogates fixed and instead *measure* the mismatch mechanism, action-response roughness induced by temporal resolution, and resolve it by *role separation* rather than by changing the surrogate's training objective.

The shaped reward of Section III-D has a recognizable form here, and the relationship is worth stating plainly. Model-ensemble methods keep several dynamics models and use their disagreement to stop the policy exploiting any one model's errors [47], and offline model-based RL makes the device explicit by subtracting a model-uncertainty term from the reward [48]. Our censor is structurally the same instrument. Two things differ, and they are what this paper contributes. First, disagreement is measured not inside an ensemble of like models fitted to a common objective but between two deliberately unlike ones, a smooth control-oriented black box and a calibrated physics-informed twin, so the penalty carries physical information a homogeneous ensemble lacks. Second, the models are assigned roles rather than averaged: the twin never enters the rollout, so policy search still runs over the smooth dynamics that make it tractable. The mechanism is thus an instance of a known family; what is new is the role assignment and its measured effect on a building-control benchmark.

D. TRANSFER ACROSS BUILDINGS

Transfer-oriented HVAC work reduces the cost of adapting DRL controllers across buildings, zones and climates [49]–[51]. These lines rarely separate two questions kept distinct here: whether the surrogate recalibrates to the target physics, and whether a frozen controller stays deployable through a new actuator interface, which the version-locked hydronic transfer study of Section IV-F answers separately.

TABLE 1. Experimental taxonomy. The central fidelity–utility test of Table 4 compares BB hourly, matched BB, calibrated GB and the hybrid; raw GB appears only in the predictive comparison of Table 3, to quantify the calibration gain. Hierarchical DRL, multi-objective RL and hydronic transfer are boundary tests.

Label	Step	Role in the study
BB hourly	3600 s	Smooth rollout baseline and positive control.
matched BB	900 s	Temporal-resolution ablation isolating whether BB's utility comes from smoothing rather than black-box structure.
raw GB	900 s	Uncalibrated predictive reference quantifying the calibration gain.
calib. GB	900 s	High-fidelity physical twin; direct-training negative control.
hybrid	900 s	Remedy: BB rollout dynamics plus frozen GB as a model-disagreement reward censor.

III. MATERIALS AND METHODS

This section defines the experimental objects: the BOPTEST control problem, the surrogate taxonomy, the hybrid backend, the controller families and the evaluation metrics. Table 1 fixes the naming used throughout. Only the first four backends are part of the central fidelity–utility test.

A. TEST CASE AND RUNTIME ENVIRONMENT

The plant is the BOPTEST `bestest_air` emulator [10], [11], a single-zone office of $6\times8\times2.7$ m ($48\,\mathrm{m}^2$) whose envelope follows the ASHRAE Standard 140 BESTEST Case 900 definition, with heavyweight concrete-block walls, an insulated slab and $12\,\mathrm{m}^2$ of south glazing. Conditioning is an idealized four-pipe fan-coil unit: a gas boiler of constant efficiency 0.9 feeds the heating coil and a chiller of constant COP 3.0 the cooling coil, with a variable-speed supply fan. The zone is a two-person office occupied 08:00–18:00 with light internal gains and occupied comfort setpoints of $21/24\,^\circ\mathrm{C}$, under Denver-Stapleton TMY weather.

Control is applied through the BOPTEST Run-Time Environment, a Dockerized Modelica emulator with embedded baseline local-loop and supervisory controllers, a disturbance forecaster and a KPI calculator, all exposed through a RESTful HTTP API [12]. The agent does not replace the plant: it *overwrites* one control input, the supply-air temperature setpoint, while the remaining embedded loops run at baseline. At each $900\,\mathrm{s}$ control step it reads measurements (zone temperature, CO_2, power) and disturbance forecasts, forms the observation, writes the setpoint through the BOPTEST overwrite pair, and advances the emulator one step; the KPIs are then computed by the runtime from the resulting live trajectory rather than from the surrogate.

The transfer study holds out three single-zone-like hydronic test cases with different heat sources, distribution paths and actuator interfaces (Table 2). They form an increasing-difficulty ladder away from the source case and share the same single-zone first-order energy balance.

B. PROBLEM FORMULATION AND EVALUATION METRICS

Because the controller never observes the full latent state, control is a partially observable Markov decision process. The agent receives an observation o_t and selects $a_t \sim \pi(\cdot \mid o_t)$ so as to maximize the expected discounted return [52]

$$J(\pi) = \mathbb{E}_\pi\left[\sum_{t=0}^{T-1}\gamma^t r_t\right], \qquad r_t = -\big(w_\mathrm{c}c_t + w_\mathrm{e}e_t\big), \quad (1)$$

with c_t a thermal-discomfort term, e_t the heating energy over the step, and $w_\mathrm{c}, w_\mathrm{e}$ fixed trade-off weights. This scalarized comfort–energy objective follows the standard multi-objective treatment of HVAC control, where preferences are represented by explicit weights rather than hidden inside a single black-box reward [23]–[25]. The comfort component is piecewise-linear with an asymmetric, ambient-dependent slope, steeper for cold-weather undershoot and fixed a priori from this physical asymmetry rather than tuned to any backend, plus a small in-band bonus; the energy component is proportional to heating power.

Because HVAC control depends on forecasts, a purely instantaneous five-dimensional observation is insufficient [37]: the canonical interface is a 17-dimensional observation comprising physical states, cyclic time features, ambient forecasts and previous actions. The agent acts through a single normalized supply-temperature command,

$$a_t \in [-1,1], \qquad T_{\mathrm{sup},t} = 18 + \tfrac{a_t+1}{2}\,(35-18) \quad [^\circ\mathrm{C}], \quad (2)$$

with a $1.0\,^\circ\mathrm{C}$ deadband, a per-step rate limit and a 21–$24\,^\circ\mathrm{C}$ comfort band. The surrogate transition is *trained* on a two-dimensional command, the supply-temperature action together with a normalized fan setting, both excited only while the offline corpus is generated; during every RL rollout and every live evaluation the fan is held at its BOPTEST baseline and only the supply-temperature setpoint is commanded.

For evaluation, as opposed to training, deployed controllers are scored with the duration-and-severity maintenance score of Wang et al. [16],

$$m_s = r_\mathrm{time} + r_\mathrm{sev}, \quad (3)$$

where $r_\mathrm{time} = N_\mathrm{viol}/N \in [0,1]$ is the fraction of control steps in which the zone temperature leaves the comfort band $[T_\mathrm{lo}, T_\mathrm{hi}]$ and

$$r_\mathrm{sev} = \max_t \max\big((T_{\mathrm{zone},t} - T_\mathrm{hi})/T_\mathrm{hi},\ (T_\mathrm{lo} - T_{\mathrm{zone},t})/T_\mathrm{lo},\ 0\big)$$

is the largest single-step band exceedance normalized by the band bound. Both terms are dimensionless and lower is better; because $r_\mathrm{time} \le 1$, the threshold $m_s > 1$ flags a controller that is either out of band for a large fraction of the horizon or commits a severe worst-case excursion, the regime treated here as a failed deployment. Separating the shaped training reward (1) from this unshaped safety score follows safe-RL practice [38]. Predictive fidelity is reported as the multi-step rollout RMSE of the zone temperature; control utility as m_s, the comfort-violation fraction and heating energy, always measured on the live emulator and never on the surrogate.

TABLE 2. BOPTEST test cases used in this study [10]. All are single-zone or single-zone-like and heating-capable. The first is the source case on which the surrogates are built; the other three are held out for the transfer study of Section IV-F.

Test case	Type	Heat source / emitter	Climate
`bestest_air` (source)	Office, $48\,\mathrm{m}^2$	Gas boiler + chiller / 4-pipe fan-coil	Denver
`bestest_hydronic`	Residential, $48\,\mathrm{m}^2$	Gas water heater / radiator ($5\,\mathrm{kW}$)	Brussels
`bestest_hydronic_heat_pump`	Residential, $192\,\mathrm{m}^2$	Air-to-water heat pump ($15\,\mathrm{kW}$) / floor loop	Brussels
`singlezone_commercial_hydronic`	Commercial, $8500\,\mathrm{m}^2$	District heating / AHU + radiator circuit	Copenhagen

C. SURROGATE BACKENDS

All surrogates approximate the same discrete-time transition $s_{t+1} = \Phi(s_t, a_t, d_t)$ and are exposed behind a single adapter, so that only the internal dynamics change across comparisons, never the control interface or the observation map. Each is fit to BOPTEST-generated transition tuples and used as a *deterministic point-transition* map rather than a learned conditional distribution, which is appropriate where the zone's conditional dynamics are effectively deterministic given the observed drivers, but does not represent transition uncertainty.

a: Control-oriented black-box surrogate (BB)

BB is a compact black-box neural predictor (8482 parameters) of a temperature increment and a power head, $\widehat{T}_{t+1} = T_t + f_{\theta,T}(x_t)$ and $\widehat{P}_{t+1} = f_{\theta,P}(x_t)$ with $\widehat{P} \geq 0$, trained by a multi-horizon (1-, 2-, 4-step) supervised rollout objective with a physical-consistency penalty. Its design goal is not long-horizon accuracy but a *smooth, well-conditioned* response surface for policy-gradient search. Its input vector is deliberately low-dimensional: thermal state, outdoor disturbance, cyclic time embedding and the direct supply-temperature action [17]. The canonical checkpoint is the early-stopped optimum of a 215-epoch run with held-out one-step $R^2 = 0.991$. Its transition carries *no* explicit time step, so the increment learned on the canonical $3600\,\mathrm{s}$ corpus is applied unchanged at the $900\,\mathrm{s}$ runtime step.

b: Physics-informed grey-box surrogate (GB)

GB replaces the black box with a first-principles lumped-capacity zone balance whose unmodeled heat flow is supplied by a neural residual head,

$$C_{\mathrm{zon}} \frac{\mathrm{d}T_{\mathrm{zone}}}{\mathrm{d}t} = \dot{Q}(x;\phi), \qquad \dot{Q}(x;\phi) = q_{\mathrm{scale}}\, q_\phi(x), \quad (4)$$

with $q_{\mathrm{scale}} = 3000\,\mathrm{W}$, integrated by a clipped explicit Euler step at the control period, and with C_{zon} positively reparameterized and bounded below by $5 \times 10^4\,\mathrm{J/K}$. Embedding this conservation law makes the model physics-informed in the sense of [8], [9], [29], [30], so C_{zon} is physically interpretable, although the analysis treats it as an *effective* capacitance that may absorb unmodeled fast dynamics rather than a direct measurement of zone air capacitance. Because C_{zon} is a property of the building rather than of the data-generating policy, it is expected to remain meaningful across test cases, which is what makes the inverse identification below worthwhile.

The canonical GB is calibrated by a staged inverse identification [5]: Stage A aligns telemetry and removes latency and bias; Stage B identifies C_{zon} as a maximum-a-posteriori estimate under a weak Gaussian prior, restricted to excitation-rich transients where the sensitivity $\partial\widehat{T}/\partial C$ is large, since quasi-steady windows would merely pull the estimate back to the prior; and Stage C refines the neural residual heads with C_{zon} frozen. The identified $C_{\mathrm{zon}} = 4.413 \times 10^5\,\mathrm{J/K}$ is a $+5.1\%$ update over the physical prior, within one standard error of the Laplace/Fisher interval, and corresponds to about $439\,\mathrm{kg}$ of air, the expected order for this zone; a supply-temperature sweep at fixed state gives the physically correct sign in 100% of 400 sampled states.

A third backend, matched-resolution BB, is the *same* black-box architecture retrained on a $900\,\mathrm{s}$ corpus. It exists solely to isolate the model-class confound in Section IV-B. Retraining on that corpus turns out to break the model's control channel (Section IV-A), so the backend is carried in two forms: as trained, and with a monotonicity constraint that restores directional validity. The constraint penalizes, at jittered states, any predicted temperature decrease in response to an increased supply-temperature command; it touches no other part of the recipe.

D. ROLE-SEPARATED HYBRID TRAINING ENVIRONMENT

The hybrid backend is not a new RL algorithm. It is an application-specific role-separation strategy grounded in three existing lines of work: model-based and world-model training in learned environments [45], [46], physics-informed and grey-box surrogate modeling [5], [8], [30], and safe-RL reward shaping [16], [38]. The contribution is the allocation of roles inside surrogate-trained HVAC control (Fig. 2): BB supplies the smooth rollout dynamics, while a *frozen* GB acts only as a per-step reward-shaping censor. Training uses the shaped reward

$$\tilde{r}_t = r_t - \lambda_T\, \delta_{T,t} - \lambda_P\, \delta_{P,t}, \quad (5)$$

where the disagreement terms are the per-step L_1 distances between the one-step predictions of the rollout surrogate f_θ (BB) and the frozen twin g_ϕ (GB), separately for the zone-temperature and HVAC power channels,

$$\delta_{\bullet,t} = \left| f_{\theta,\bullet}(s_t, a_t, d_t) - g_{\phi,\bullet}(s_t, a_t, d_t) \right|,$$

with no squaring, normalization or clipping. The two censor weights differ in scale because they multiply a temperature residual in °C and a power residual in W; for the thermostatic controller $\lambda_T = 0.10$ and $\lambda_P = 5 \times 10^{-5}$.

Crucially, g_ϕ is evaluated *forward only* and its outputs are detached, so the censor enters the *reward* alone, never the policy loss, and never the rollout dynamics, which remain those of BB. Because a policy-gradient method estimates its gradient from sampled returns rather than by differentiating the environment, this role split does not preserve any environment-to-policy gradient; what it preserves is the smooth trajectory-and-return distribution of BB, and its favorable optimization conditioning, while penalizing state–action regions on which the frozen twin disagrees with the rollout. In other words, GB does not replace BB, form an accuracy-oriented ensemble, or introduce a second policy-loss term. The penalty measures disagreement between two imperfect surrogates, not violation of a conservation law or of a physically feasible set; it is therefore a *model-disagreement* censor, physics-informed through GB's structure but not a hard physical-feasibility filter.

E. CONTROLLER FAMILIES AND PROTOCOL

All policies are optimized with proximal policy optimization (PPO) [53] using generalized advantage estimation [54] and the Stable-Baselines3 implementation [55]. Shared settings are learning rate 3×10^{-4}, discount factor $\gamma = 0.99$ (an effective horizon of about 25 h at the 900 s step) and ten optimization epochs per rollout; the families differ in rollout length, minibatch size and total timestep budget. The recipe is evaluated across three controller families of increasing structure: a *thermostatic* PPO agent on the shared forecast-augmented observation; a *hierarchical* DRL controller (HDRL) that decomposes control across timescales in the options/feudal tradition [21], [22] as recently applied to year-round HVAC operation [20]; and a *preference-conditioned* multi-objective controller (MORL) [23], [26].

Policies are trained on their surrogate backend and then evaluated *zero-shot* in closed loop on the live emulator over two targeted 14-day windows: a January peak-heat window with daily-mean ambient $-24.4\,^\circ$C and a February typical-heat window at $2.4\,^\circ$C. Because a policy trained on a surrogate is deployed on a different, live system, the central methodological risk is distribution shift [39]; the evaluation therefore always reports the live-emulator score. Each hypothesis, pass threshold and recalibration adapter was prespecified *before* the corresponding runs.

a: Implementation

Surrogates use three corpora generated by scripted excitation: a canonical BB corpus (51 200 rows at 1 h), a 15-minute exploration corpus (48 384 rows at 900 s) and a GB calibration corpus (10 744 rows at 900 s); the deliberate one-hour/fifteen-minute pairing is the instrument behind the matched-corpus ablation. All controllers are trained in Stable-Baselines3 2.1.0 under Python 3.11.9. Experiments ran on a laptop workstation (6-core CPU, 32 GB RAM, laptop-class GPU) with the emulator in a Docker container. Behind one HTTP contract sit four interchangeable backends (live BOPTEST, BB, GB, hybrid), so only the dynamics backend changes across the surrogate-versus-live comparison; the surrogate backends run 85.0–220.2 times faster than the live loop, so a 5×10^6-step policy-optimization run costs about 47 minutes of environment time against roughly 66 hours on the live runtime. Canonical KPI tables use seed 42; the four central backends additionally carry an $N = 3$ seed-robustness audit over seeds $\{42, 43, 44\}$, MORL is evaluated over five fixed seeds, and the HDRL censor-weight sweep is replicated over the same three seeds at each of its four λ_T settings.

IV. RESULTS AND DISCUSSION

The experiments answer four questions in sequence. *First*, which surrogate is the more faithful predictor? The calibrated GB, by a wide margin (Section IV-A). *Second*, does that higher fidelity make it the better RL training environment? No: used directly, both GB and a matched-resolution BB collapse on the live runtime (Section IV-B), which is the paper's central causal test. *Third*, why do the high-fidelity backends collapse, and how is control recovered (Sections IV-C–IV-D)? *Fourth*, how far does the result transfer (Sections IV-E–IV-F)?

A. PREDICTIVE FIDELITY: THE CALIBRATED TWIN IS THE BETTER MODEL

Predictive accuracy is reported as the multi-step rollout $\mathrm{RMSE}_T(h)$ of the per-step error $\widehat{T}_i(t+h) - T_i(t+h)$, together with the engineering-tolerance error CDF and its 95th-percentile tail; the coefficient of determination is taken against the held-out mean, so a *negative* 24 h R^2 means the rollout is worse than predicting that mean. Over 24 h rollouts the calibrated GB reaches $\mathrm{RMSE}_T = 0.644\,^\circ$C, against $1.466\,^\circ$C for the uncalibrated GB and $1.557\,^\circ$C for the canonical hourly BB (Fig. 3). The hourly BB has a *negative* 24 h R^2, worse than the historical mean over long horizons and worse than a naive persistence forecast at 24 h (1.557 against $0.969\,^\circ$C), yet it remains a usable control surrogate because policy-gradient search needs smooth local gradients, not long-horizon realism. Calibrated GB instead keeps a positive 24 h R^2, bounds its tail (24 h P95 $1.207\,^\circ$C) and beats persistence at every horizon. The episode-level diagnostic matters because a low aggregate error can hide a single unstable rollout: the calibrated GB stays below the $1\,^\circ$C engineering reference in all eight held-out episodes, with a best episode of $0.486\,^\circ$C and a worst of $0.964\,^\circ$C, which supports predictive validity as a repeated-episode result rather than a single-window artifact.

Because the original BB and the canonical GB artifacts use different data resolutions, a corpus-matched retraining step is required: without it, a reviewer could correctly attribute the GB gain to the 15-minute corpus rather than to Stage A/B/C calibration. Retraining the same black-box architecture on the matched corpus brings it to $0.876\,^\circ$C (Table 3). Of the total $0.913\,^\circ$C improvement from hourly BB to calibrated GB, $0.681\,^\circ$C (74.6%) is the corpus shift and $0.232\,^\circ$C (25.4%) the staged calibration; an alternative split through the uncalibrated GB backbone gives 9.9%/90.1%, the two differing because corpus and architecture interact non-additively. The

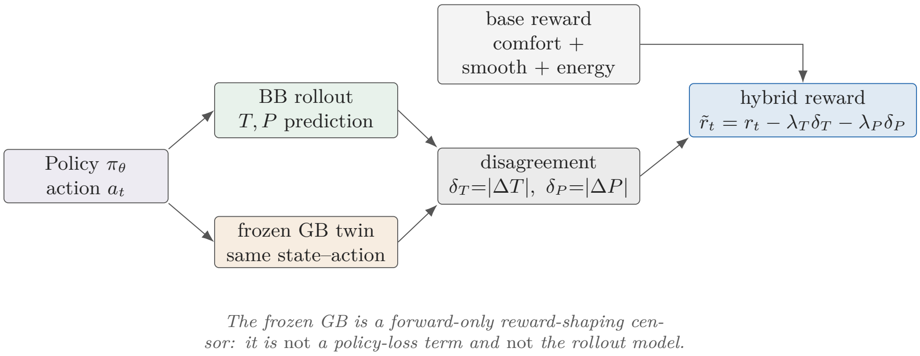

FIGURE 2. **Hybrid reward-shaping mechanism. The black-box surrogate (BB) supplies the smooth rollout dynamics; the frozen grey-box surrogate (GB) is evaluated forward-only on the same state–action pair and enters *only* as a per-step reward penalty on BB–GB disagreement, Eq. (5). GB is neither the rollout model nor a policy-loss term.**

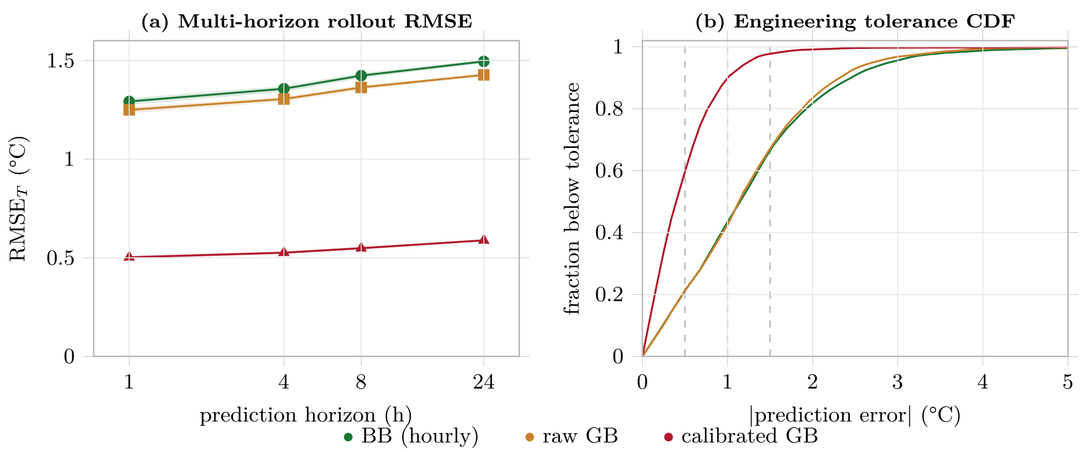

FIGURE 3. **Multi-horizon predictive validity, residual spread and engineering error CDFs from BOPTEST rollouts. The calibrated grey-box twin is the better predictor at every horizon; the coarse black-box surrogate degrades beyond a few hours yet remains the usable training environment.**

TABLE 3. **Corpus-matched comparison. The second row separates the corpus-resolution contribution from the Stage A/B/C calibration contribution. Step is in seconds; *Calib.* names the applied inverse-calibration stages, which do not apply to the black-box variants. *Sign* is the fraction of 400 sampled states in which raising the supply-temperature command does not lower the predicted zone temperature. The matched-resolution retraining improves rollout error and inverts the control channel; the monotonicity-constrained variant (third row) improves both.**

Variant	Step	Calib.	24 h RMSE_T	Sign
BB hourly	3600	—	1.557	100%
BB matched	900	—	0.876	**9.8%**
raw GB	900	off	1.466	100%
calibrated GB	900	A+B+C	0.644	100%

calibration claim is therefore positive but bounded, and it is reported symmetrically rather than as a single number.

a: Rollout error does not detect a lost control channel

Predictive error is not the only property the corpus change moves. Auditing directional validity shows the matched-resolution BB inverting the supply-temperature channel in about nine states out of ten, while its rollout error improves. The inversion is not a defective draw: four independent trainings, differing only in seed, give 3.2–14.0 % valid states with median response −1.14–−0.84 °C, and over the same four draws the 24 h rollout error *improves* to 0.696–0.876 °C. The metric the surrogate literature reports, and the one this study deliberately chose over one-step error, is blind to it.

The defect is not confined to one input. The same audit applied to the fan command finds the unconstrained matched-resolution model valid in only 2.2% of states with hot supply air, against 98.5% for the hourly BB and 98.8% for the calibrated GB. Constraining the supply channel during training raises its own validity to 100% at a *lower* rollout error (0.745–0.817 °C over three seeds) but leaves the fan channel at 20.8%, so the constraint relocates the defect rather than removing it. Two conclusions follow. Directional validity is something a surrogate can lose while its predictive score improves, and it must be audited on *every* actuated input, because a controller will find whichever channel is wrong: policies and planners trained on these models drive the fan to its minimum and deliver almost no heat. Both checks cost seconds and run before any controller is trained.

The runtime benchmark closes the predictive-validation step: the calibrated GB is $114.2\times$ faster than the live HTTP loop and the hybrid reference $85.0\times$, so the surrogate family is not only more accurate after calibration but fast enough to support repeated policy-training experiments.

B. HIGHER PREDICTIVE FIDELITY DOES NOT IMPLY TRAINING UTILITY

Used directly as PPO rollout environments, both the more accurate GB and a matched-resolution BB collapse on the live runtime, whereas the less accurate hourly BB trains a usable controller. This is the central causal test, and it compares four training backends under the same thermostatic PPO controller. Direct GB is a *negative control*, not a failed implementation: it asks whether the highest-fidelity predictive surrogate can be used directly as the policy rollout environment.

Table 4 sets predictive fidelity against live control utility, Fig. 4 shows the measured evidence chain and Fig. 5 plots fidelity against utility directly. The result contradicts the working assumption of the surrogate-training paradigm. Pure BB PPO is already usable ($m_s = 0.073$ peak, 0.095 typical). Direct GB PPO fails catastrophically despite its superior predictive fidelity: live violation reaches 77.1% on the peak window and 82.4% on the typical window, with closed-loop RMSE above $4.3\,°\mathrm{C}$. This is not a grey-box artifact: the matched-resolution BB, that is the *same* black-box architecture made strictly more accurate, collapses as well ($m_s = 1.142$/1.211, violation 85.6/91.4%). That backend is directionally inverted on both actuated inputs (Section IV-A), so it cannot by itself isolate model class: a controller trained on it fails for a reason that has nothing to do with temporal resolution. We attempted to repair it with a training constraint on the supply channel; the repaired model is more accurate and valid in that channel, but its fan channel remains inverted and the controller trained on it exploits that instead, so the matched-resolution point does not isolate the mechanism either way. The mechanism argument below therefore rests on the calibrated GB, which is directionally sound on both inputs, and on the rescaling control, which preserves response signs by construction. The hybrid backend resolves the conflict, reaching $m_s = 0.087$ on the peak window and 0.041 on the typical window with violation below 5% on both and lower energy than pure BB on the peak window.

The decomposition $m_s = r_{\mathrm{time}} + r_{\mathrm{sev}}$ exposes the failure structure directly (Fig. 6). Direct GB's peak score is dominated by $r_{\mathrm{time}} \approx 0.77$, that is out of band 77% of the time, plus a large worst-case severity $r_{\mathrm{sev}} \approx 0.28$; the hybrid's peak score of 0.087 is almost entirely a small severity term with $r_{\mathrm{time}} < 0.05$. The hybrid therefore does not merely lower the average error, it removes the sustained band departures that dominate the direct-GB score.

Retrained on three fixed seeds, each backend keeps its regime: pure BB and the hybrid hold every seed below the 5% violation bar on both windows, while the matched-resolution BB collapses on every seed ($m_s = 1.14$/1.21, standard deviation below 0.001) and direct GB collapses on every seed ($m_s = 1.07 \pm 0.02$ / 1.13 ± 0.03). The usable-versus-collapse verdicts are therefore seed-stable, and the single-seed peak-window edge of pure BB over the hybrid does not persist across seeds.

C. MECHANISM: TEMPORAL COARSENESS, ROUGHNESS AND SATURATION

The high-fidelity backends collapse because their action-response surfaces are rougher, which drives near bang-bang saturation and live failure. The effect is not confined to policy-gradient learning. A receding-horizon MPC planning through the same surrogates reproduces the same ordering: it reaches $m_s = 0.547$/0.869 on the coarse hourly BB, the least accurate backend, and 1.051/1.369 on the calibrated GB, the most accurate one, landing within 0.005 of the learner on the peak window (1.046). Whatever the surrogates do to a controller, they do it to a planner as well, so the operative interaction is between the response surface and gradient-based optimization through the model rather than between the response surface and policy-gradient search specifically. One caveat bounds this: the planner optimizes its action sequence with a first-order method through the differentiable surrogate, so it shares any pathology gradients have on a rough surface. The comparison therefore separates gradient-based optimization from other synthesis routes, not planning from learning; a derivative-free planner over the same backends is the test that would separate those, and is left to future work. The mechanism is a measured chain from surrogate response geometry to policy behavior to live outcome (Fig. 7). Sweeping each backend's one-step map $\widehat{T}(a_0)$ over a grid of states and computing a scale-free *relative roughness*, mean curvature divided by mean slope, orders the backends cleanly with usability: the usable hourly BB is smoothest (0.17), while the matched-resolution BB and GB are $9.4\times$ and $7.9\times$ rougher. Both collapsing backends saturate the actuator, placing extreme actions in temperature-error regimes the live building does not support, in 100% of steps against 24% for the usable controller. The matched-resolution backends are excluded from this comparison because their directional defects give their controllers a separate and sufficient reason to fail: the unconstrained model's policy holds the minimum supply command (10th and 90th percentiles both $18.0\,°\mathrm{C}$) while the zone sits below band, and the supply-constrained model's policy instead holds the fan near zero (mean 0.201, mean power $182\,\mathrm{W}$ against $959\,\mathrm{W}$ for the usable controller). Both are the behavior a controller learns when the model tells it that acting is counterproductive. The time-domain signature is direct GB saturating and leaving the comfort band while the coarse BB and the hybrid stay in band (Fig. 8).

Relative roughness is necessary but not sufficient. A constant Δt rescaling preserves the curvature-to-slope ratio, so a Δt-rescaled hourly BB has the *same* relative roughness of 0.17 as the usable hourly BB, yet it also collapses on every seed ($m_s = 1.19/1.22$). The operative surrogate property is therefore the coarse per-step increment magnitude, of which

TABLE 4. Central fidelity–utility paradox. Each backend trains the same thermostatic PPO controller, evaluated zero-shot in closed loop on the live BOPTEST runtime over the peak/typical 14-day windows. Predictive fidelity (24 h rollout RMSE, $^\circ$C) is set against live control utility (m_s, unusable when $m_s > 1$, and comfort violation). Point values are the canonical seed-42 runs, dispersions are standard deviations over seeds $\{42, 43, 44\}$; the usable-versus-collapse verdict spans more than an order of magnitude and is not a seed effect. Averaged over the three seeds the hybrid leads pure BB on *both* windows (0.060 against 0.072 peak, 0.014 against 0.046 typical), so the canonical-seed peak ordering here is a single-seed artifact. The hybrid rolls out on BB and has no independent predictive RMSE.

			m_s (canonical seed $\pm$ seed s.d.)		Violation (%)	
Backend	Step (s)	24 h RMSE	peak	typical	peak	typical
BB hourly	3600	1.557	0.073 ± 0.019	0.095 ± 0.047	1.5	4.4
matched BB	900	0.876	1.142 ± 0.000	1.211 ± 0.000	85.6	91.4
calibrated GB	900	0.644	1.046 ± 0.022	1.102 ± 0.026	77.1	82.4
hybrid	900	—	0.087 ± 0.026	0.041 ± 0.024	4.7	2.4

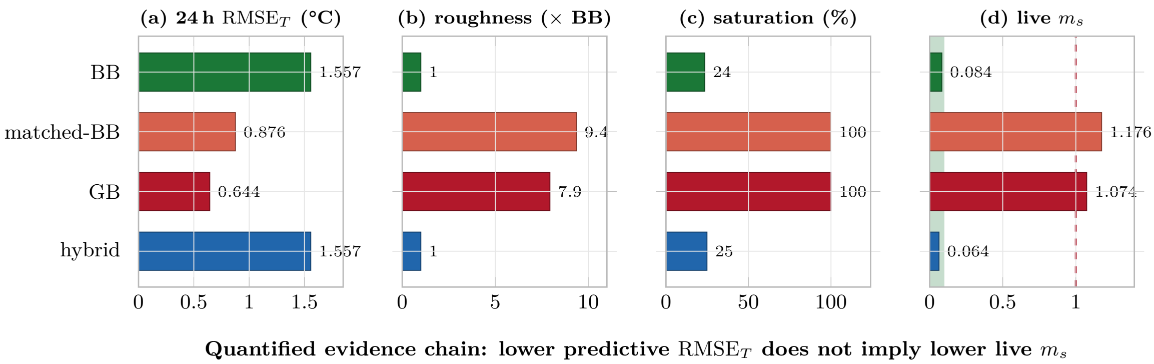

FIGURE 4. The fidelity–utility paradox at a glance. For the four thermostatic-PPO backends the measured chain runs (A) 24 h rollout RMSE $\to$ (B) action-surface roughness $\to$ (C) policy saturation $\to$ (D) live maintenance score. The higher-fidelity backends win on (A) yet are rougher, saturate and collapse in (D), and the contradiction between (A) and (D) is the paradox. For the hybrid, panel (A) shows the BB rollout RMSE because the hybrid rolls out on BB; panel (D) reports the cross-window mean.

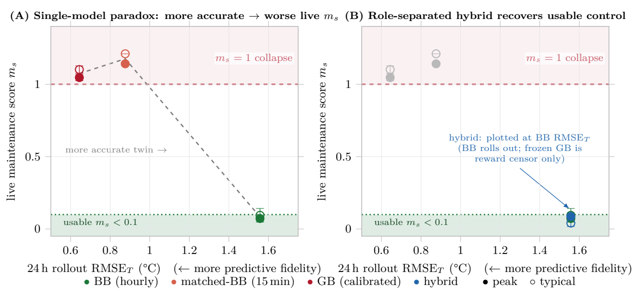

FIGURE 5. Predictive fidelity against live control utility: a lower rollout RMSE can raise the live maintenance score, and the role-separated hybrid returns to the usable region. Filled and open markers denote the peak and typical windows; error bars are ± 1 seed standard deviation over three seeds.

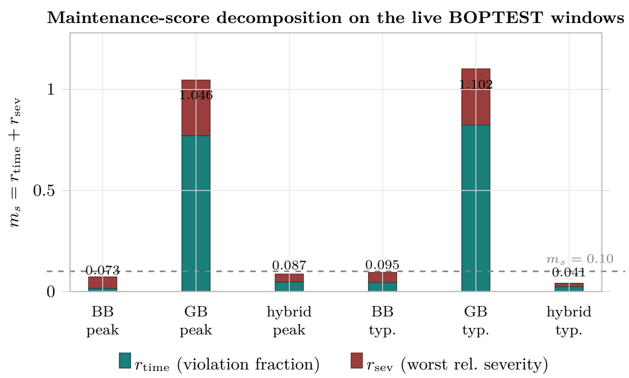

FIGURE 6. Live BOPTEST maintenance-score decomposition. Direct GB fails through sustained out-of-band time; pure BB and the hybrid keep both duration and severity small.

relative roughness measures one facet, rather than scale-free smoothness alone. This also settles which part of the coarse-versus-fine contrast is load-bearing: the hourly BB is deployed with its hourly increment applied unchanged at the 900 s step, four times the dimensionally consistent per-step change, so part of both its smoothness and its 1.557 °C error is a train/deploy timestep mismatch rather than corpus resolution alone.

Consequently the outcome across the three single-model backends is not a monotonic fidelity–utility curve but a two-level contrast: the temporally coarse backend is usable and both finer backends collapse, with a non-monotonic ordering inside the collapse region. The engineering reading is that policy-gradient search needs an optimization-friendly training landscape as well as accurate predictions, and that the two requirements are traded against one another by the control-step resolution of the training corpus. This is interpreted as an instance of model exploitation in model-based

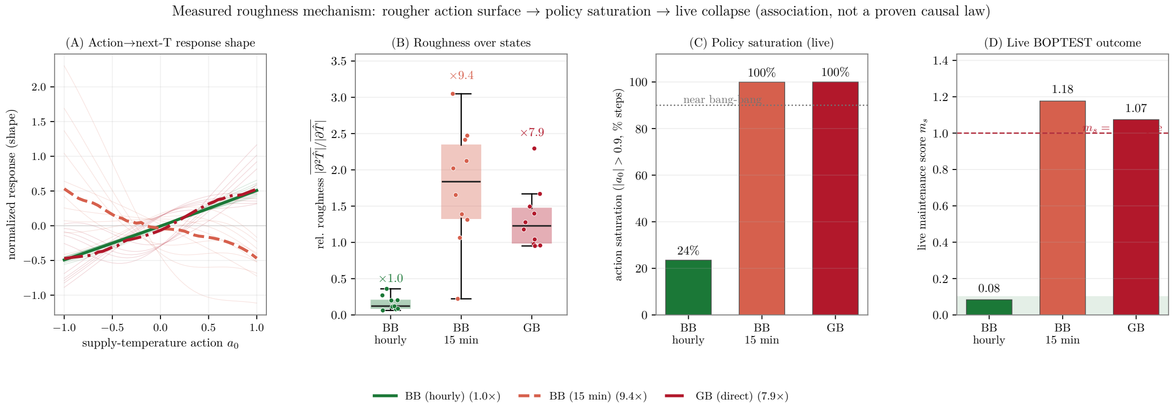

FIGURE 7. Measured roughness mechanism. The collapsing single-model backends expose rougher action–response surfaces, produce near bang-bang saturation and fail on the live emulator; the pattern is mechanistically consistent with the fidelity–utility paradox but does not by itself prove full loss-landscape causality.

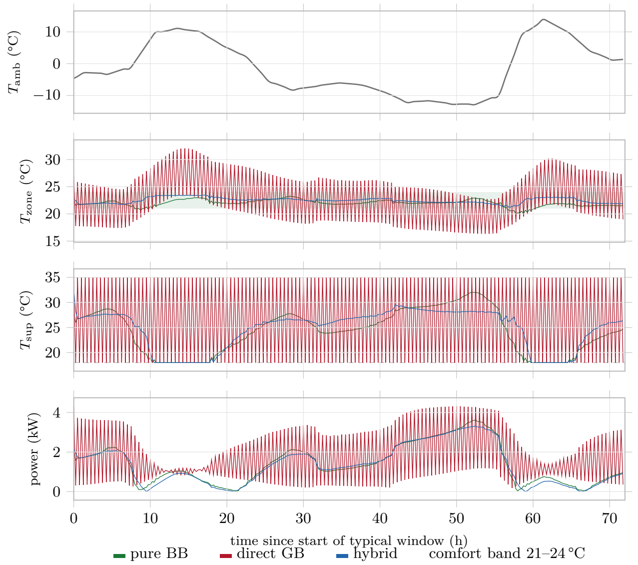

FIGURE 8. Live closed-loop behavior on the typical window: zone temperature against the comfort band, with ambient disturbance and HVAC power, for the pure-BB, direct-GB and hybrid controllers. Direct GB saturates the actuator and leaves the band; the coarse BB and the hybrid track it.

RL, a distribution-shift effect [39], [40]: the sharper response surface of the high-fidelity backends is exploited by policy-gradient search into a near bang-bang law that does not survive transfer to the live plant.

D. ROLE SEPARATION RESTORES CONTROL

Separating the two roles resolves the conflict. With BB retained as the rollout model and the frozen GB entering only through the censor of Eq. (5), the hybrid attains $m_s = 0.087$ and 0.041 on the peak and typical windows, with comfort violation below 5% on both and lower energy than pure BB on the peak window, at $85.0\times$ live throughput. On the canonical seed pure BB is marginally ahead on the peak window alone; the hybrid has the better *worst* window, and the ordering reverses on both windows once averaged over seeds (Table 4).

The canonical operating point was selected by a thermostatic sweep over $\lambda_T \in \{0.05, 0.10, 0.15\}$ at fixed $\lambda_P = 5\times10^{-5}$; the mid setting retains the energy advantage while avoiding the stronger comfort degradation seen at the weaker and stronger censor settings. Only the canonical point's live KPIs are retained as a frozen artifact: the bracketing settings served selection only and were never carried to the full two-window live protocol, so the headline results of Table 4 are not reported on data used to choose λ_T. The power weight λ_P was not swept: it was fixed a priori at the value that puts the power residual in W on the scale of the temperature residual in °C. Across the hybrid traces the mean temperature disagreement is 0.92 °C on the peak window and 1.01 °C on the typical window (95th percentiles 2.22 and 2.52 °C), and the mean power disagreement 682 W and 735 W respectively: large enough to shape learning and bounded enough to stay informative.

The hybrid should not be read as an accuracy-oriented ensemble of two predictors. If it were, GB would naturally dominate the rollout; the experiments show the opposite, since the physically richer model becomes useful only once its role is restricted to a detached model-disagreement censor, while the smoother BB remains the training substrate. A second negative control confirms this. Using GB as a policy *initializer* instead, by pretraining on GB and warm-start fine-tuning on the live runtime, is compared against an otherwise identical scratch fine-tune: it does not help, and warm-starting raises m_s by a factor of two to three on both windows. The physical twin belongs in the reward, not in the rollout dynamics and not in the weights.

Transfer-gap diagnostics confirm the same ordering from the surrogate side. Direct GB shows by far the largest surrogate-to-live mismatch, a learned bang-bang exploit whose policy sits at the opposite action bound from BOPTEST throughout, with action-gap norm $g_a \approx 2.0$, while the hybrid shows the smallest gap and stays consistent with the live runtime longest.

E. BOUNDARY TEST I: CONTROLLER FAMILIES

The preceding sections establish the paradox and its remedy within the thermostatic PPO family. Two further families test whether the recipe transfers; they are boundary tests, not additional proof of the paradox.

a: Hierarchical DRL

A seasonal hierarchy routes control to winter and summer PPO setpoint specialists that share the 17-dimensional observation and the supply-temperature action. The thermostatic-optimal censor weight $\lambda_T = 0.10$ does *not* transfer: because each specialist is already comfort-aware, the GB temperature censor over-regularizes the low-level loop and m_s degrades monotonically with λ_T, with peak violation roughly quadrupling from $6.3 \pm 1.7\%$ to $24.4 \pm 3.8\%$ (mean $\pm$ s.d. over seeds $\{42, 43, 44\}$). HDRL is best at $\lambda_T = 0$, keeping only the power channel. The correct censor strength is therefore family-specific, and a single "best" regularization strength cannot be reported without naming its controller family. The sweep is seed-replicated at each of its four settings: m_s rises from 0.182 ± 0.017 to 0.442 ± 0.033 on the peak window and from 0.245 ± 0.017 to 0.570 ± 0.077 on the typical window, an endpoint separation of 10.0 and 5.8 pooled seed standard deviations with disjoint ± 1 s.d. bands. The ordering is monotone in the seed means at every step, but the response saturates: the $0.05 \rightarrow 0.10$ increment is only 1.4 pooled standard deviations on both windows and is not resolved above seed noise, so the supported statement is that the censor degrades hierarchical control across the swept range, not that every increment is separately resolvable. The evidence against censor-weight universality is therefore two independently swept families, thermostatic PPO and HDRL, and on the HDRL side it is now seed-replicated; H3 is accordingly reported as *falsified*.

b: Multi-objective RL

A four-stage MORL pipeline (surrogate pretraining, preference adaptation, live fine-tuning, yearly evaluation) on the power-only hybrid backend is viable only once the observation is widened from five to 17 dimensions: the five-dimensional interface fails ($\mathrm{RMSE}_T = 2.721\,^\circ\mathrm{C}$, 37.8% violation, $m_s = 0.680$) while the forecast-augmented 17-dimensional vector recovers a usable policy ($\mathrm{RMSE}_T = 0.720\,^\circ\mathrm{C}$, 4.9%, $m_s = 0.099$). The bottleneck is observation geometry, not the reward scalarization, consistent with forecast-augmented RL HVAC control [37]. MORL yields a usable comfort–energy Pareto front, but an $N = 5$ seed extension turns the canonical operating points into wide distributions (coefficient of variation 0.42–0.61) that fail a pre-specified stability test: promising in the mean, but not deployment-stable without explicit stabilization.

c: Synthesis against the PI reference

The BOPTEST built-in PI controller is a reproducible reference, comfort-poor but energy-frugal, with 12-month means $\mathrm{RMSE}_T = 3.40\,^\circ\mathrm{C}$, 63.6% violation and $m_s = 0.910$. All evaluated learned agents reach m_s well below PI, dominating it on comfort at comparable or lower energy. The cross-family maintenance-score decomposition makes the pattern explicit: the failures, direct GB and PI, carry a large time-in-violation term $r_{\mathrm{time}} \gtrsim 0.6$, PI by chronic under-heating and direct GB by saturated overshoot, whereas the usable controllers keep both terms small.

F. BOUNDARY TEST II: CROSS-BUILDING TRANSFER

Transfer is decomposed into two components, frozen-policy transfer and surrogate-recalibration transfer, which prevents a misleading binary conclusion. The claim is deliberately bounded: three pre-selected single-zone hydronic cases, documented actuator adapters, pre-specified recalibration, and target-test-case controller fine-tuning explicitly excluded. A pre-specified per-test-case adapter maps the frozen policy's supply-temperature command to each target's BOPTEST override inputs, committed and smoke-tested for directional validity before any run. The three targets differ in heat source, distribution path and actuator set but share the same single-zone first-order energy balance, so transfer changes the *delivery path* and the actuator interface while C_{zon} remains the shared lumped coordinate that Stage B re-identifies. The controller-side pass threshold is normalized by each test case's built-in PI baseline, $\tau = 1.25\, m_{s,\mathrm{PI}}$, with the factor pre-specified rather than chosen post hoc; energy is reported on a separate axis, because the commercial case passes the m_s bound while consuming substantially more energy than PI.

Table 5 contains two stories. The surrogate component transfers strongly: full Stage A/B/C recalibration improves rollout RMSE by 60.2–87.8% on all three targets, and the re-identified capacitance clusters at $\rho_C = 1.918 \pm 0.032$ times the source value across an order-of-magnitude change in floor area, read as an *effective* single-zone RC signature rather than a claimed physical invariant. The frozen controller does not transfer uniformly: both residential cases fail the $1.25\times$-PI threshold while *saving* energy, and the commercial stretch case passes the threshold at a 35.3% energy penalty. Mean delivered power exposes the regime: about 0.7–2.8 kW for the residential cases, conduction and slow-hydronic dominated, but 76.9 kW for the commercial case, ventilation and AHU dominated, about $28\times$ the heat-pump target, which is the physical reason the commercial controller passes comfort yet overshoots energy. No frozen-policy transfer is deployment-ready (Fig. 9). This component-level boundary is consistent with building heterogeneity requiring local adaptation rather than zero-shot reuse [50], [51]. For practice, the transferable asset is the calibration pipeline, not the frozen control law.

G. LIMITATIONS

Evaluation is simulation-only on a single-zone emulator, with multi-zone and richer topologies untested and transfer limited to a hydronic family. Several caveats bound the central claim. First, the fidelity–utility relationship is established at two directionally sound operating points, hourly BB and calibrated GB, together with the rescaling control, rather

TABLE 5. **Transferability matrix. $m_{s,\mathrm{RL}}$ and $m_{s,\mathrm{PI}}$ are the maintenance scores of the frozen RL controller and the built-in PI baseline; the pass threshold is $\tau = 1.25\,m_{s,\mathrm{PI}}$ and the controller verdict is PASS when $m_{s,\mathrm{RL}} \leq \tau$; ´E = 100($E_{\mathrm{RL}} - E_{\mathrm{PI}}$)/$E_{\mathrm{PI}}$; raw and full RMSE ($^\circ$C) are 24 h rollout errors before and after staged recalibration, with relative gain G; ρ_C is the re-identified capacitance ratio against the source value 4.413×10^5 J/K. † threshold pass with an energy penalty.**

Test case	$m_{s,\mathrm{RL}}$	$m_{s,\mathrm{PI}}$	τ	Ctrl.	$\Delta E\%$	Raw RMSE	Full RMSE	$G\%$	ρ_C
hydronic heat pump	0.665	0.464	0.579	FAIL	−7.3	1.421	0.565	60.2	1.892
hydronic (boiler)	0.976	0.750	0.938	FAIL	−5.8	2.666	0.335	87.4	1.954
commercial hydronic	0.431	0.628	0.785	PASS†	35.3	1.952	0.238	87.8	1.909

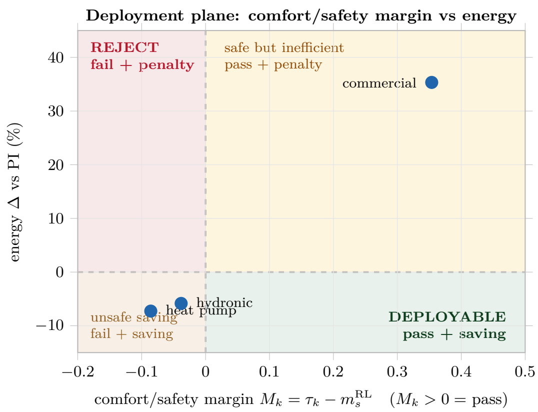

FIGURE 9. **Comfort–energy deployment plane. Residential hydronic cases save energy but fail the comfort threshold; the commercial case passes the threshold with a 35.3% energy penalty. No frozen-policy transfer is deployment-ready.**

than as a continuous curve. The matched-resolution backend is reported as a predictive-metric failure rather than as a mechanism data point, because both of its actuated channels are directionally defective and a training constraint on one relocates the defect to the other; an intermediate-resolution sweep on a surrogate audited across all inputs is left to future work. Second, controller utility is established against two references, BOPTEST's built-in PI ($m_s = 0.910$) and a pre-registered receding-horizon MPC planning on the same surrogates. The MPC is the stronger of the two, reaching $m_s = 0.547/0.869$ on the coarse BB backend, but it is a fixed, untuned configuration rather than a tuned Guideline-36 or industrial MPC: its comfort violation is still 34.5% and 55.6% on the two windows. It was deliberately not tuned against m_s, the metric it is scored on, so it should be read as a stronger reference than PI rather than as a state-of-the-art controller. Third, the surrogate power channel is reproduced less accurately than temperature, with a calibrated power-head mean absolute error of 482 W, about 9% of $P_{\max} = 5500$ W; the reported energy KPIs and the transfer ΔE are live-versus-live measurements that do not inherit this error, but the weaker power channel limits predictive-fidelity claims about the power head and adds noise to the training-time energy reward. Fourth, the evidence against H3 now rests on a seed-replicated four-point HDRL sweep ($N = 3$ per setting), but the sweep density remains non-uniform across families: two measured endpoints for thermostatic PPO and a single adopted configuration for MORL. What it establishes is that the thermostatic optimum does not transfer to the hierarchical family, not a fully swept per-family optimum for every family; and within the HDRL sweep the response saturates above $\lambda_T = 0.05$, so the ranking of the two largest weights is not resolved above seed noise. Finally, the mechanism is *measured* statically on each surrogate's one-step action map and corroborated by the Δt-rescaling control, which preserves response signs by construction, rather than by the matched-resolution ablation, whose backend is directionally defective. A loss-landscape analysis along the training trajectory and generality to off-policy algorithms remain open.

V. CONCLUSION

Surrogate models for reinforcement-learning HVAC control are selected by predictive error. This paper shows that criterion to be wrong and replaces it. On the BOPTEST benchmark the most accurate surrogate in the study, a calibrated physics-informed twin with a 24 h rollout error of 0.644 °C, trains a controller that leaves the comfort band for more than 77% of the horizon on every seed, while the least accurate one (1.557 °C) trains a usable controller; making the black-box surrogate more accurate by refining its resolution (0.876 °C) makes it unusable too, so the effect is not attributable to model class. H1 is therefore falsified.

The finer-resolution surrogate additionally exposes a second failure of the predictive criterion. Retraining on the finer corpus inverts the model's response to the supply-temperature command in about 90% of sampled states, reproducibly over four draws, while rollout error improves in all of them; the policy trained on such a model holds the minimum supply command while the building sits below band. The fan command is inverted as well, and a training constraint that repairs the supply channel leaves the fan channel broken, with the resulting controller holding the fan near zero instead. Predictive error therefore fails as a selection criterion in two distinct ways, and the second one must be tested on every actuated input rather than on the model as a whole. The matched-resolution backend is accordingly reported as evidence about the criterion, not about the mechanism.

What does predict training utility is the magnitude of the surrogate's per-step increment. A rescaling control that holds the scale-free roughness of the action-response surface fixed while changing only the increment reproduces the collapse, which separates the operative variable from the smoothness with which it is otherwise confounded. Because the increment is a property of the surrogate alone, it can be evaluated before a controller is trained, which makes it usable as a selection criterion rather than only as an explanation after the fact.

Role-separated surrogate training turns that finding into an architecture. Rather than compromising between smoothness and fidelity inside one model, the two requirements are assigned to two: a smooth black-box surrogate carries the rollout dynamics and a frozen physics-informed twin enters only as a per-step model-disagreement reward censor. This gives the best worst-window maintenance score of the study ($m_s = 0.087$ peak and 0.041 typical, against 0.073/0.095 for the best single-model backend) with comfort violation below 5% on both windows and an $85\times$ training speed-up, and H2 is supported. Two negative controls, warm-starting from the twin and using it as an accuracy-oriented partner, both fail, which locates the gain in the role assignment itself.

The remaining two hypotheses are not supported, and reporting them is part of the contribution. The censor weight does not transfer across controller families: on a sweep replicated over three seeds at every setting, the weight that is optimal for the thermostatic family degrades the hierarchical one across the whole swept range, by 10.0 and 5.8 pooled seed standard deviations at the endpoints. H3 is falsified, and no single regularization strength can be quoted without naming the family it belongs to. Transfer to related buildings resolves into a component-level boundary: the inverse-calibration pipeline ports, with rollout error falling by 60.2–87.8% and the zone capacitance re-identifying at a near-uniform ratio across an order-of-magnitude change in floor area, whereas the frozen control policy does not transfer zero-shot.

For practice the implication is concrete: a surrogate intended as an RL training environment should be qualified on its per-step increment and on the smoothness of its action–response surface, and validated on downstream closed-loop utility, not on predictive error alone. Future work follows from the limitations of Section IV-G: multi-zone and mixed-topology buildings, a loss-landscape analysis along the training trajectory, a validated rather than auxiliary heating-power channel, seed replication for the preference-conditioned family, and deployment on a physical building.

A receding-horizon MPC planning through the same surrogates was added as a second reference, pre-registered as H5b and reported whether favorable or not. It is the stronger of the two baselines, reaching $m_s = 0.547$/0.869 against the emulator PI's 0.910, and it does not approach the role-separated controller. It also reproduces the inversion: 0.547/0.869 on the least accurate backend against 1.051/1.369 on the most accurate one, within 0.005 of the learner on the peak window. H5b, which predicted that a planner would escape the effect, is therefore falsified, and the scope of the mechanism is wider than the RL framing implied. The planner is itself a first-order method through the differentiable surrogate, so this separates gradient-based optimization from other synthesis routes rather than planning from learning; a derivative-free planner is the test that would separate those.

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REFERENCES

[1] K. Al Sayed, A. Boodi, R. Sadeghian Broujeny, and K. Beddiar, "Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review," *J. Build. Eng.*, vol. 95, p. 110085, 2024.

[2] A. T. Nguyen, D. H. Pham, B. L. Oo, M. Santamouris, Y. Ahn, and B. T. H. Lim, "Modelling building HVAC control strategies using a deep reinforcement learning approach," *Energy Build.*, vol. 310, p. 114065, 2024.

[3] D. Hou and R. Evins, "A protocol for developing and evaluating neural network–based surrogate models and its application to building energy prediction," *Renew. Sustain. Energy Rev.*, vol. 193, p. 114283, 2024.

[4] M. Mshragi and I. Petri, "FastML-GA: FPGA-accelerated machine learning for real-time energy HVAC optimization in buildings," *Neural Comput. Appl.*, vol. 38, p. 25, 2026.

[5] P. Bacher and H. Madsen, "Identifying suitable models for the heat dynamics of buildings," *Energy Build.*, vol. 43, no. 7, pp. 1511–1522, 2011.

[6] F. Bünning, B. Huber, P. Heer, A. Aboudonia, and J. Lygeros, "Experimental demonstration of data predictive control for energy optimization and thermal comfort in buildings," *Energy Build.*, vol. 211, p. 109792, 2020.

[7] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.

[8] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.

[9] S. Hedayat, T. Ziarati, and M. Manganelli, "A physics-informed reinforcement learning framework for HVAC optimization: Thermodynamically-constrained deep deterministic policy gradients with simulation-based validation," *Energies*, vol. 18, no. 23, p. 6310, 2025.

[10] D. Blum, J. Arroyo, S. Huang, J. Drgoňa, F. Jorissen, H. T. Walnum, Y. Chen, K. Benne, D. Vrabie, M. Wetter, and L. Helsen, "Building optimization testing framework (BOPTEST) for simulation-based benchmarking of control strategies in buildings," *J. Build. Perform. Simul.*, vol. 14, no. 5, pp. 586–610, 2021.

[11] M. Wetter, W. Zuo, T. S. Nouidui, and X. Pang, "Modelica buildings library," *J. Build. Perform. Simul.*, vol. 7, no. 4, pp. 253–270, 2014.

[12] J. Arroyo, C. Manna, F. Spiessens, and L. Helsen, "An Open-AI gym environment for the building optimization testing (BOPTEST) framework," in *Proc. 17th IBPSA Conf. Build. Simul.*, vol. 17, 2021, pp. 175–182.

[13] J. R. Vázquez-Canteli, J. Kämpf, G. Henze, and Z. Nagy, "CityLearn v1.0: An OpenAI gym environment for demand response with deep reinforcement learning," in *Proc. ACM Int. Conf. Syst. Energy-Efficient Build. Cities Transp. (BuildSys)*, 2019, pp. 356–357.

[14] J. R. Vázquez-Canteli and Z. Nagy, "Reinforcement learning for demand response: A review of algorithms and modeling techniques," *Appl. Energy*, vol. 235, pp. 1072–1089, 2019.

[15] S. Savino, G. Razzano, M. Pagone, C. Novara, and A. Capozzoli, "Deploying deep reinforcement learning for low-level HVAC control in multi-zone buildings: A comparative study with ASHRAE G36 sequences," *Energy Build.*, vol. 348, p. 116456, 2025.

[16] X. Wang, P. Wang, R. Huang, X. Zhu, J. Arroyo, and N. Li, "Safe deep reinforcement learning for building energy management," *Appl. Energy*, vol. 377, p. 124328, 2025.

[17] L. Sun, Z. Hu, M. Mae, and T. Imaizumi, "Individual room air-conditioning control in high-insulation residential building during winter: A deep reinforcement learning-based control model for reducing energy consumption," *Energy Build.*, vol. 323, p. 114799, 2024.

[18] J. Deng, X. Wang, and F. Meng, "Multi-agent deep reinforcement learning for smart building energy management with chance constraints," *Energy Build.*, vol. 331, p. 115408, 2025.
[19] B. S. Alotaibi, "Context-aware smart energy management system: A reinforcement learning and IoT-based framework for enhancing energy efficiency and thermal comfort in sustainable buildings," *Energy Build.*, vol. 340, p. 115804, 2025.
[20] C. Liao, S. Miyata, M. Qu, and Y. Akashi, "Year-round operational optimization of HVAC systems using hierarchical deep reinforcement learning for enhancing indoor air quality and reducing energy consumption," *Appl. Energy*, vol. 390, p. 125816, 2025.
[21] P.-L. Bacon, J. Harb, and D. Precup, "The option-critic architecture," in *Proc. AAAI Conf. Artif. Intell.*, vol. 31, 2017.
[22] O. Nachum, S. Gu, H. Lee, and S. Levine, "Data-efficient hierarchical reinforcement learning," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
[23] R. Yang, X. Sun, and K. Narasimhan, "A generalized algorithm for multi-objective reinforcement learning and policy adaptation," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
[24] D. M. Roijers, P. Vamplew, S. Whiteson, and R. Dazeley, "A survey of multi-objective sequential decision-making," *J. Artif. Intell. Res.*, vol. 48, pp. 67–113, 2013.
[25] C. F. Hayes, R. Rădulescu, E. Bargiacchi, J. Källström, M. Macfarlane, M. Reymond, T. Verstraeten *et al.*, "A practical guide to multi-objective reinforcement learning and planning," *Auton. Agent. Multi-Agent Syst.*, vol. 36, no. 1, p. 26, 2022.
[26] W. Byeon, G. Park, J. Chae, A. Leshem, and Y. Sung, "Multi-objective reinforcement learning with max-min criterion: A game-theoretic approach," in *Adv. Neural Inf. Process. Syst.*, 2025.
[27] T. Berthou, P. Stabat, R. Salvazet, and D. Marchio, "Development and validation of a gray box model to predict thermal behavior of occupied office buildings," *Energy Build.*, vol. 74, pp. 91–100, 2014.
[28] D. Picard, J. Drgoňa, M. Kvasnica, and L. Helsen, "Impact of the controller model complexity on model predictive control performance for buildings," *Energy Build.*, vol. 152, pp. 739–751, 2017.
[29] A. Karpatne, G. Atluri, J. H. Faghmous, M. Steinbach, A. Banerjee, A. Ganguly, S. Shekhar, N. Samatova, and V. Kumar, "Theory-guided data science: a new paradigm for scientific discovery from data," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 10, pp. 2318–2331, 2017.
[30] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Comput. Surv.*, vol. 55, no. 4, p. 66, 2022.
[31] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, "Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators," *Nat. Mach. Intell.*, vol. 3, no. 3, pp. 218–229, 2021.
[32] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, "Fourier neural operator for parametric partial differential equations," in *Int. Conf. Learn. Represent.*, 2021.
[33] P. Benner, S. Gugercin, and K. Willcox, "A survey of projection-based model reduction methods for parametric dynamical systems," *SIAM Rev.*, vol. 57, no. 4, pp. 483–531, 2015.
[34] D. R. Jones, M. Schonlau, and W. J. Welch, "Efficient global optimization of expensive black-box functions," *J. Glob. Optim.*, vol. 13, no. 4, pp. 455–492, 1998.
[35] W. Mayor, J. Obando-Ceron, A. Courville, and P. S. Castro, "The impact of on-policy parallelized data collection on deep reinforcement learning networks," in *Proc. 42nd Int. Conf. Mach. Learn. (ICML)*, vol. 267, 2025.
[36] M.-B. Radac and D.-P. Chirla, "Near real-time online reinforcement learning with synchronous or asynchronous updates," *Sci. Rep.*, vol. 15, p. 17158, 2025.
[37] Y. Gao, S. Shi, S. Miyata, and Y. Akashi, "Successful application of predictive information in deep reinforcement learning control: A case study based on an office building HVAC system," *Energy*, vol. 291, p. 130344, 2024.
[38] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," *J. Mach. Learn. Res.*, vol. 16, no. 1, pp. 1437–1480, 2015.
[39] J. Quiñonero Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, *Dataset Shift in Machine Learning*. Cambridge, MA: MIT Press, 2009.
[40] A. Riahi Samani, X. Zhao, and F. Chen, "Distribution shift, generalization and OOD challenge in offline reinforcement learning: a comprehensive survey," *Neural Comput. Appl.*, vol. 38, p. 150, 2026.
[41] N. O. Lambert, B. Amos, O. Yadan, and R. Calandra, "Objective mismatch in model-based reinforcement learning," in *Proc. 2nd Conf. Learn. Dyn. Control (L4DC)*, vol. 120, 2020, pp. 761–770.
[42] M. Janner, J. Fu, M. Zhang, and S. Levine, "When to trust your model: Model-based policy optimization," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
[43] A.-m. Farahmand, A. M. S. Barreto, and D. Nikovski, "Value-aware loss function for model-based reinforcement learning," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, vol. 54, 2017, pp. 1486–1494.
[44] W. Zhao, J. P. n. Queralta, and T. Westerlund, "Sim-to-real transfer in deep reinforcement learning for robotics: a survey," in *IEEE Symp. Ser. Comput. Intell.*, 2020, pp. 737–744.
[45] R. S. Sutton, "Dyna, an integrated architecture for learning, planning, and reacting," *ACM SIGART Bull.*, vol. 2, no. 4, pp. 160–163, 1991.
[46] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, vol. 640, pp. 647–653, 2025.
[47] T. Kurutach, I. Clavera, Y. Duan, A. Tamar, and P. Abbeel, "Model-ensemble trust-region policy optimization," in *Int. Conf. Learn. Represent.*, 2018.
[48] T. Yu, G. Thomas, L. Yu, S. Ermon, J. Y. Zou, S. Levine, C. Finn, and T. Ma, "MOPO: model-based offline policy optimization," in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020.
[49] F. Hou, J. C. P. Cheng, H. H. L. Kwok, and J. Ma, "Multi-source transfer learning method for enhancing the deployment of deep reinforcement learning in multi-zone building HVAC control," *Energy Build.*, vol. 322, p. 114696, 2024.
[50] D. Coraci, S. Brandi, T. Hong, and A. Capozzoli, "An innovative heterogeneous transfer learning framework to enhance the scalability of deep reinforcement learning controllers in buildings with integrated energy systems," *Build. Simul.*, vol. 17, no. 5, pp. 739–770, 2024.
[51] D. Coraci, A. Silvestri, G. Razzano, D. Fop, S. Brandi, E. Borkowski, T. Hong, A. Schlueter, and A. Capozzoli, "A scalable approach for real-world implementation of deep reinforcement learning controllers in buildings based on online transfer learning: The HiLo case study," *Energy Build.*, vol. 329, p. 115254, 2025.
[52] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.
[53] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
[54] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Int. Conf. Learn. Represent.*, 2016.
[55] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.

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